

Instructions:

- **Due Date:** May 19th, 11:59 PM. Submit all materials to Blackboard as (i) a single PDF report and (ii) a compressed (.zip) archive containing all supporting files. This deadline is firm; late submissions will not be accepted.
- **Team Policy:** This is a team project consisting of three members per team. Once teams are assigned, you may **not** discuss or collaborate with students outside your team.
- **Report Format:** The project report must be neat, clear, and professionally prepared. All reports must be written in L^AT_EX. Reports prepared in Word or handwritten submissions will **not** be graded.
- **Figures and Tables:** All figures must include labeled axes and descriptive captions. All tables must include descriptive captions.
- **Document Updates:** This document may be updated for clarification or corrections. Any updates will be announced on Blackboard.

Submission Guidelines:

1. Name your report as `ENME403_S26_Project_team_<team number>.pdf`.
2. Include all figures and tables (with captions) in the report.
3. Do **not** include code in the report. Place all MATLAB scripts, Simulink models, and `.tex` files in a folder and compress it as `ENME403_S26_Project_team_<team number>.zip`. All code files must be properly referenced in the report.
4. Upload both the PDF and the archive file to Blackboard before the deadline.
5. Only **one** team member should submit the files on behalf of the team.

Academic Integrity:

- Include the following statement in your report:
"We have neither received nor given aid to anyone outside our team during this project."
Sign and print all team members' names.
- Reports with significant similarity will be referred to the UMBC Academic Conduct Committee.

Author Contributions:

Include a brief author contribution statement. For example:

- Author 1: Methodology, Software
- Author 2: Data curation, Writing – Original draft
- Author 3: Visualization, Investigation, Validation

Stabilization and Inversion of a Reaction-Wheel Pendulum

A planar pendulum with a reaction wheel, shown in Figure 1, consists of a rigid body $\mathcal{B}$ of mass m and a reaction wheel $\mathcal{W}$ of mass M mounted at its end. The pendulum is attached to a frictionless pin joint at the point w . The center of mass of the pendulum body is located at a distance ℓ from the pivot, and the reaction wheel is located at a distance L from the pivot.

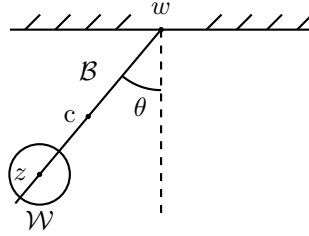


Figure 1: Planar pendulum with a reaction wheel.

Let θ denote the angular position of the pendulum, and let ψ denote the angular position of the reaction wheel, both measured from the vertical. An internal torque τ_m is generated by the motor between the pendulum body and the reaction wheel.

The equations of motion of the system are

$$(J_{\mathcal{B}/w} + ML^2)\ddot{\theta} + J_{\mathcal{W}/z}\ddot{\psi} + (m\ell + ML)g \sin \theta = 0, \quad (1)$$

$$J_{\mathcal{W}/z}\ddot{\psi} = \tau_m, \quad (2)$$

where $J_{\mathcal{B}/w}$ is the moment of inertia of the pendulum body about the pivot, and $J_{\mathcal{W}/z}$ is the moment of inertia of the reaction wheel about its axis.

The equations of motion can be written compactly as

$$\mathcal{M}\ddot{q} + \mathcal{D}(q, \dot{q}) = T, \quad (3)$$

where $q \triangleq \begin{bmatrix} \theta \\ \psi \end{bmatrix}$ and

$$\mathcal{M} \triangleq \begin{bmatrix} J_{\mathcal{B}/w} + ML^2 & J_{\mathcal{W}/z} \\ 0 & J_{\mathcal{W}/z} \end{bmatrix}, \quad \mathcal{D}(q, \dot{q}) \triangleq \begin{bmatrix} (m\ell + ML)g \sin \theta \\ 0 \end{bmatrix}, \quad T \triangleq \begin{bmatrix} 0 \\ \tau_m \end{bmatrix}. \quad (4)$$

The torque τ_m is generated by a DC motor. Let i denote the armature current and v denote the motor terminal voltage. The motor dynamics are given by

$$\tau_m = K_t i, \quad (5)$$

$$L_a \dot{i} + R_a i + K_e(\dot{\psi} - \dot{\theta}) = v, \quad (6)$$

where K_t is the torque constant, K_e is the back-emf constant, R_a is the armature resistance, and L_a is the armature inductance. The physical properties of the system are shown in Table 1.

In this project, you will design a feedback control system to stabilize the pendulum in the inverted orientation using voltage as the input.

Parameter	Value
Mass of pendulum body, m	0.5 kg
Mass of reaction wheel, M	0.2 kg
Distance to pendulum body center of mass, ℓ	0.15 m
Distance to reaction wheel center, L	0.20 m
Pendulum body inertia about pivot, $J_{B/w}$	0.02 kg m ²
Reaction wheel inertia about its axis, $J_{W/z}$	0.005 kg m ²
Motor torque constant, K_t	0.05 N m/A
Motor back-emf constant, K_e	0.05 V s/rad
Armature resistance, R_a	2 Ω
Armature inductance, L_a	0.5 mH
Acceleration due to gravity, g	9.81 m/s ²

Table 1: Physical properties of the reaction-wheel pendulum.

Part 1: Full-state Feedback Control

1. **State-Space Formulation.** Define the state x and input u . Write down the nonlinear system (1), (2), (6) in the state-space form

$$\dot{x} = f(x, u). \quad (7)$$

Hint: Define the state vector as

$$x \triangleq \begin{bmatrix} \theta \\ \dot{\theta} \\ \omega \\ i \end{bmatrix}, \quad (8)$$

where $\omega \triangleq \dot{\psi}$ is the angular velocity of the reaction wheel. Note that the dynamics do not depend explicitly on ψ , so it is not included as a state. Let the input u be the motor voltage v .

2. **Equilibrium Solution.** Find all equilibrium solutions of the nonlinear system (7). Equilibrium solution is the pair $(x_{\text{eq}}, u_{\text{eq}})$ such that $f(x_{\text{eq}}, u_{\text{eq}}) = 0$.
3. **Linearization.** Linearize the nonlinear system (7) about all equilibria obtained in the previous step. At each equilibrium, classify the linearized system as asymptotically stable, Lyapunov stable, semi-stable, or unstable.
4. **Stabilization of the Upright Position.** At the unstable equilibrium $\theta = \pi$, design a stabilizing full-state feedback controller. You may use either the `place` command to assign the closed-loop eigenvalues or LQR to compute a stabilizing gain.

Simulate the closed-loop system in Simulink to validate your design. The required architecture is shown in Figure 2. Since the linearization is valid only in a local neighborhood of $\theta = \pi$, choose the initial condition $\theta(0)$ close to the upright position.

Using your designed controller, determine the range of $\theta(0)$ for which the system converges to the upright equilibrium. Assume that all other state components start at 0.

Hints/Suggestions:

- (a) The system dynamics (7) are provided as a Simulink block in the [shared Matlab Drive](#).
- (b) For LQR, start with $R_1 = I_4$ and $R_2 = 10^{-2}$.
- (c) Use initial conditions of the form $x(0) = [\pi + w_1 \quad w_2 \quad w_3 \quad w_4]^T$, where w_i are small perturbations.

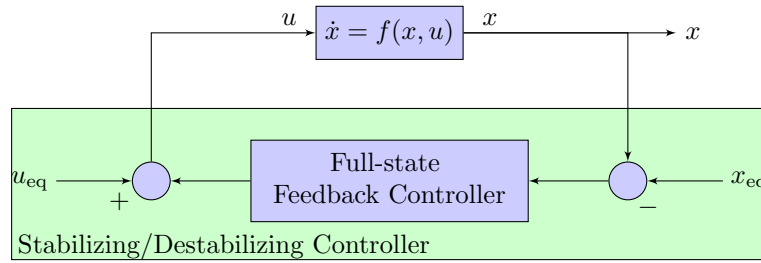


Figure 2: Full-state feedback stabilization/destabilization of an equilibrium point.

5. **Destabilization of the Downward Position.** At the stable equilibrium $\theta = 0$, design a full-state feedback controller that renders the equilibrium unstable. Use the `place` command to assign the closed-loop eigenvalues. Simulate the closed-loop system in Simulink to validate your design.

Since the linearization is valid only in a local neighborhood of $\theta = 0$, the state x may remain close to the destabilized equilibrium even when the closed-loop eigenvalues lie in the open right-half plane. Consequently, the control input u may not provide sufficient excitation to drive the pendulum toward the upright position, where the stabilizing controller can take over.

A purely real unstable pole may move the pendulum away from $\theta = 0$ only through a slow non-oscillatory drift, which may not generate the repeated swings needed for inversion. To produce a swing-up motion, it is often more effective to assign a lightly unstable complex-conjugate pair. This creates oscillations whose amplitude grows with time, allowing the pendulum to build enough energy to approach the upright position. At the same time, the remaining modes should be kept stable to prevent the controller from excessively exciting the entire system.

To guide the pole selection, note that near $\theta = 0$, the pendulum dynamics are approximately

$$(J_{B/w} + ML^2)\ddot{\theta} + (m\ell + ML)g\theta \approx 0.$$

Thus, the small-angle natural frequency of the pendulum mode is approximately

$$\omega_n \approx \sqrt{\frac{(m\ell + ML)g}{J_{B/w} + ML^2}}.$$

Using the parameters in Table 1, $\omega_n \approx 6.35$ rad/s. Therefore, a reasonable choice is to place the unstable complex pair near this frequency,

$$\lambda_{1,2} = \sigma \pm j\omega_d, \quad \omega_d \approx 6 \text{ rad/s}, \quad \sigma > 0 \text{ small}.$$

The real part σ should be chosen small enough to produce gradually growing oscillations rather than an excessively aggressive response. The remaining eigenvalues should be placed in the open left-half plane, with one corresponding to the electrical dynamics typically placed farther to the left.

Hints/Suggestions:

- A fully destabilizing controller places all closed-loop eigenvalues in the open right-half plane, which may drive the state too far from the equilibrium for the stabilizing controller to recover. Since the objective is to move the system away from $\theta = 0$ without excessively exciting all modes, placing only one oscillatory mode in the open right-half plane is often sufficient.
- A useful starting point is to place a lightly unstable complex-conjugate pair near the pendulum mode, and keep the remaining modes stable. For example,

$$\lambda_{1,2} = 0.2 \pm 6j, \quad \lambda_3 = -15, \quad \lambda_4 = -1000.$$

You may need to tune the real part of the unstable pair to obtain oscillations that grow fast enough for swing-up, but not so fast that the stabilizing controller cannot recover the pendulum

near $\theta = \pi$. If the pendulum drifts away from $\theta = 0$ without developing noticeable oscillations, the unstable poles are likely too close to the real axis. If the response becomes too aggressive, reduce the positive real part of the unstable pair.

6. **Switching Control.** Since the pendulum starts near $\theta = 0$ and the objective is to invert it, combine the destabilizing and stabilizing controllers designed in the previous steps to construct a *switching controller*. The switching controller consists of a destabilizing *swing-up* controller and a stabilizing *catching* controller.

- The swing-up controller is active near $\theta = 0$ and drives the pendulum away from the downward equilibrium. Thus, it effectively destabilizes the $\theta = 0$ equilibrium.
- The catching controller is active when the pendulum approaches $\theta = \pi$ and stabilizes the upright equilibrium. The catching controller is activated when $\theta \in \mathcal{C}$, where the set $\mathcal{C}$ contains $\theta = \pi$. For example, $\mathcal{C} = (3\pi/4, 5\pi/4)$.

In this step, combine the two controllers to implement the switching controller shown in Figure 3.

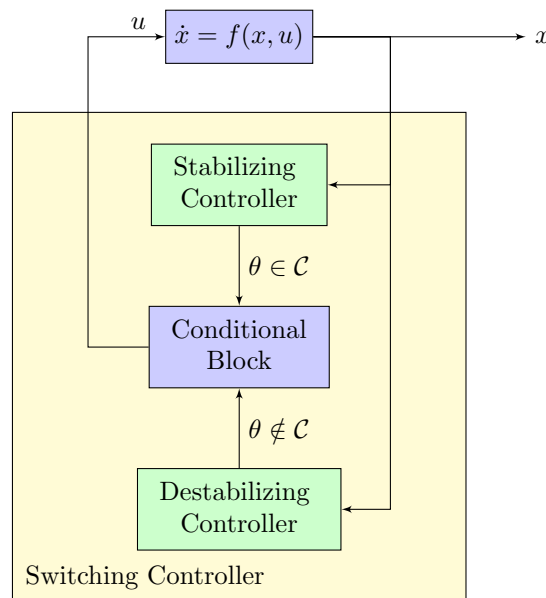


Figure 3: Switching controller block diagram to swing up and stabilize the pendulum in the up position.

Part 2: Controller Analysis

7. **Gain & Phase Margin.** At the upright equilibrium $\theta = \pi$, analyze the robustness of your controller using frequency-domain tools. Using the linearized model about $\theta = \pi$,

$$\dot{x} = Ax + Bu,$$

and the state-feedback law

$$u = Kx,$$

form the loop transfer function

$$L(s) = K(sI - A)^{-1}B.$$

Compute the gain margin and phase margin of $L(s)$ using Bode or Nyquist methods. Comment on whether the margins indicate a robust or fragile design.

8. **Robustness to Parameter Uncertainty.** Next, to numerically test the robustness of your controller, scale the pendulum mass m by a scalar factor α . The new pendulum mass is thus αm , where $\alpha \in [0.1, 2]$. For each value of α , simulate the nonlinear closed-loop system using your switching full-state feedback controller. Evaluate the effect of parameter variation on

- stability (does the system remain stable?),
- transient response (rise time, overshoot),
- control effort.

Comment on the sensitivity of your controller to modeling errors. In particular, discuss whether the controller maintains acceptable performance across the full range of α , and identify any values of α for which performance degrades significantly or instability occurs.

9. **Effect of Measurement Noise.** To study the effect of measurement noise, assume that the measured angle is corrupted by noise, that is,

$$x_{\text{meas}} = x + we_1, \quad (9)$$

where w is zero-mean Gaussian noise and e_1 is the first column of the 4×4 identity matrix. Simulate the noise using the Random Number block in Simulink, and vary its variance from 10^{-6} to 10^{-1} .

Using your switching full-state feedback controller with x_{meas} in place of x , simulate the closed-loop system and analyze

- degradation in stabilization performance,
- behavior of the control signal $u(t)$ (noise amplification),
- sensitivity of the closed-loop system to increasing noise levels.

Comment on how measurement noise affects both the control effort and the ability of the controller to stabilize the pendulum. Discuss any trade-offs observed between robustness and control aggressiveness.

Grading Rubric (100 points total)

- **Part 1: Full-State Feedback Control (50 pts)**
 - Correct state-space formulation, equilibrium computation, and linearization
 - Proper stability classification of equilibria
 - Design and justification of stabilizing controller at $\theta = \pi$
 - Design and justification of destabilizing controller at $\theta = 0$
 - Implementation and validation in simulation
 - Region of attraction analysis
 - Switching controller design and performance
- **Part 2: Controller Analysis (40 pts)**
 - Gain and phase margin analysis with correct methodology
 - Proper formulation of loop transfer function using state-space representation
 - Robustness analysis under parameter variation
 - Measurement noise analysis and discussion
 - Interpretation of results (stability, performance, limitations)
- **Report Quality and Presentation (10 pts)**
 - Clarity, organization, and technical writing quality
 - Proper use of $\text{\LaTeX}$, equations, and notation
 - Quality of figures (labeled axes, captions, readability)
 - Proper referencing of code and inclusion of author contributions